

Particulate Matter Monitoring & Health Effects

Issue 2026-08-15 | Entry window 15 August 2026 | 21 records carried, 26 rejected

21RECORDS CARRIED
(26 REJECTED)**9**SUBTOPIC CLUSTERS
REPRESENTED**5**EFFECT ESTIMATES
WITH INTERVALS**7**INSTRUMENTATION /
EXPOSURE RECORDS**2**METADATA-ONLY
(NO ABSTRACT)

Signal of the day

Two large cohorts extend $PM_{2.5}$ dose–response into South Asia and sub-Saharan Africa on the same day, and both rest entirely on modelled exposure in the regions where the models are least validated.

Mandal et al. (*Kidney Int Rep*) follow 12,271 adults in Chennai and Delhi across three eGFR measurements and report -0.32 (95% CI -0.49 to -0.15) and -0.42 (-0.57 to -0.27) ml/min/1.73 m² per 5 µg/m³ of annual $PM_{2.5}$. Moro et al. (*Environ Res*) find OR 1.06 (1.02–1.11) per 1 µg/m³ for hypertension in 1,833 rural Ghanaian adults, PAF 18.6%, with no association for metabolic syndrome or any other component.

Both are worth having — renal and sub-Saharan cardiometabolic endpoints are genuine evidence gaps. But the exposure in Mandal is a 1 km national spatiotemporal model at Delhi annual means of 118–130 µg/m³, and in Moro a satellite-derived model over rural Ghana. Neither region has the reference-monitor density that calibrated the modelling approaches these estimates inherit, and neither paper reports out-of-sample model error at the exposure levels it is used at. A per-unit odds ratio of 1.06 is recoverable only if the exposure contrast is measured to better than a unit. That is a measurement claim, not an epidemiological one, and it is unaddressed in both papers — which is precisely the argument for dense low-cost networks in exactly these settings.

What changed today

- ▶ **A community-run mobile LCS campaign reports its own year-to-year model instability.** Cho et al. ran two summers of community-scientist mobile $PM_{2.5}$ monitoring in Norwood, Ohio, then fitted street-level land-use regression. Cross-validated R^2 went from 0.17–0.25 in 2020 to 0.55–0.61 in 2021 on the same streets with the same protocol. The paper reads the difference as pandemic-phase source recovery; it is equally consistent with 2020 having too little exposure contrast for LUR to resolve. Either way, publishing both years rather than the good one is the right call.
- ▶ **Combined cold-spell and heavy-pollution exposure is supra-additive for circulatory mortality.** Zhang et al. pooled 70 districts across the Beijing–Tianjin–Hebei region, cold seasons 2013–2018. CRR 1.35 (1.14–1.60) for the overlapping event against 1.14 (1.09–1.20) for cold spell alone and 1.08 (1.00–1.16) for heavy pollution alone — the last of which only touches unity. Two lagged single-exposure warnings would each have under-called the combined day.
- ▶ **A continuous HNO_3 /particulate-nitrate method at sub-0.15 µg/m³ detection limits.** Alexa et al. pair a wet diffusion denuder with a continuous aerosol sampler and chemiluminescence detection, recording at 1 s intervals, with LODs of 0.13 µg/m³ for HNO_3 and 0.10 µg/m³ for NO_3^- , validated against filter/denuder reference sampling across Brno winter and summer campaigns. Nitrate partitioning is exactly the fast process filter methods average away and low-cost optical sensors cannot see at all.
- ▶ **Twelve French sites, seven years, one harmonised PMF — and the CTM is systematically biased.** Chebaicheb et al. apply harmonised rolling PMF to ACSM organic aerosol at 12 (sub)urban sites. Primary OA factors are ≈33% of OA mass; oxygenated OA dominates everywhere; wintertime LO-OA tracks BBOA, implicating residential biomass burning in secondary formation. Comparison against CHIMERE shows systematic component-level bias. This is the reference dataset any speciated-PM epidemiology in France should be built against.
- ▶ **Nine of twenty-one records carry no health endpoint.** At 43% that share is flat against 12–14 August (46%, 41%, 43%) and the split is now a stable property of the corpus rather than a daily fluctuation. Every *Sensing* and *Exposure* record measures particles without measuring a person; every cohort record models exposure without measuring particles. Unlike 14 August, no record today measured both on the same subjects.

Corpus analytics

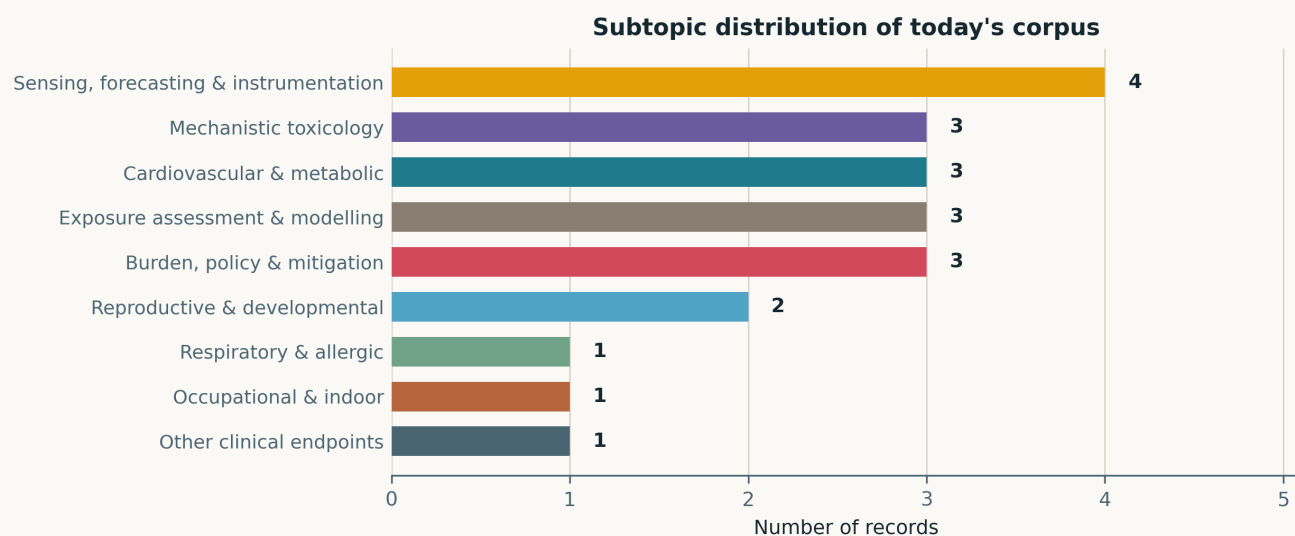


Fig. Subtopic assignment of the 21 carried records, single-label by dominant contribution. Nine of ten clusters are occupied and *Neuro / mental health* is empty for the second consecutive issue — and on 11 of the 21 issues to date, including an unbroken six-day run from 3 to 9 August, so this is the watch's least reliable cluster rather than a new gap. *Sensing, forecasting & instrumentation* leads at 4; *Mechanistic toxicology*, *Cardiovascular & metabolic*, *Exposure assessment & modelling* and *Burden, policy & mitigation* each hold 3.

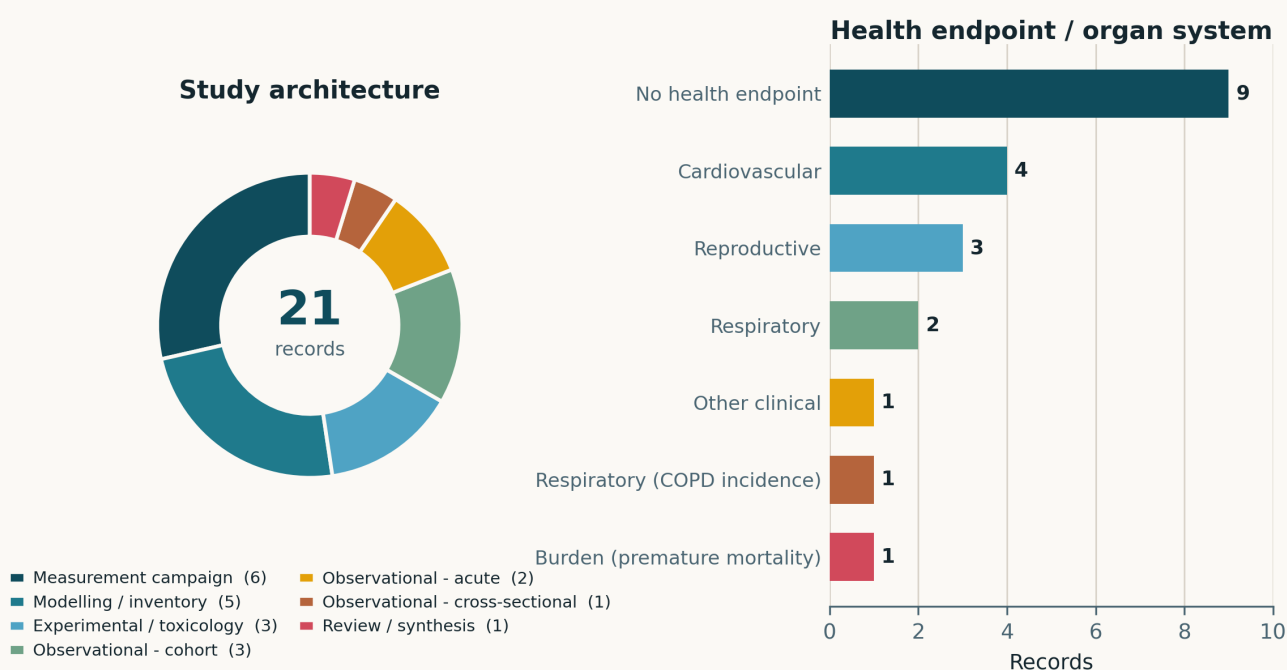


Fig. Left — study architecture. *Measurement campaign* (6) and *Modelling / inventory* (5) together take eleven of twenty-one records, the most measurement-weighted day of the week; observational designs split 3 cohort, 2 acute, 1 cross-sectional, and there is no *Trial / intervention* record at all. Right — health endpoint. **No health endpoint** is the largest bar at 9, covering the four sensing records, the three exposure/modelling records, the indoor diesel-tracer campaign and the ship-emission inventory. Cardiovascular endpoints follow at 4.

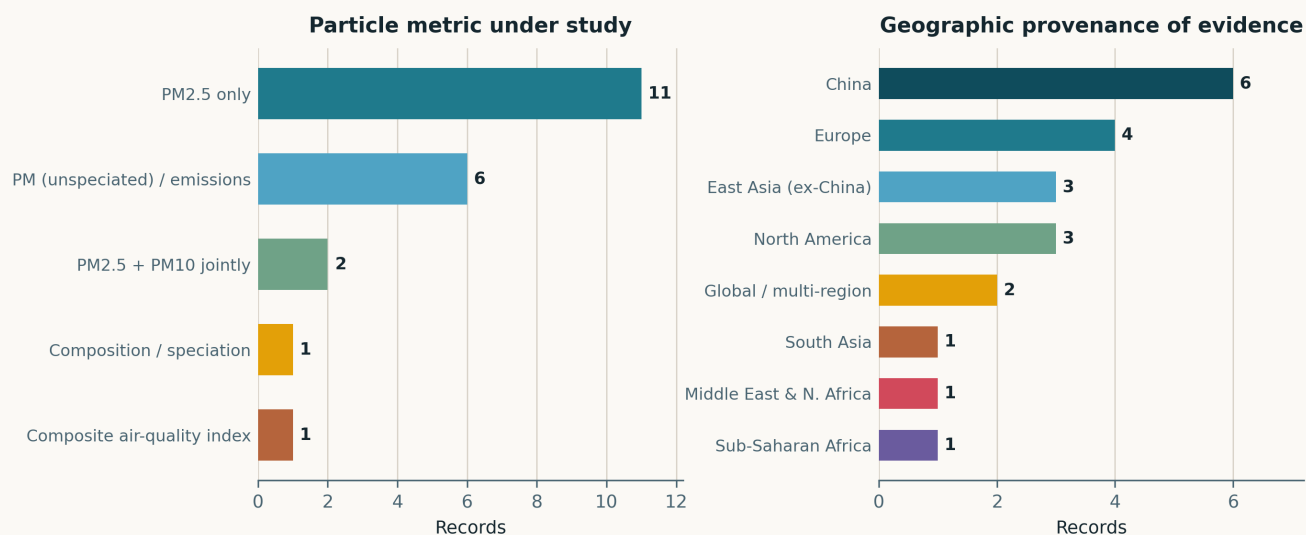


Fig. Left — particle metric. PM_{2.5}-only accounts for 11 records. Three records depart from mass: Ranjbar et al.'s nitro-PAH speciation, Chebaicheb et al.'s ACSM organic factors (binned with unspeciated/emissions), and Kafi et al., whose Sentinel-5P evaluation carries *no surface PM product at all* and is binned by the classifier as a composite index — read that slice as an absence, not a measurement. Right — geography. China contributes 6 records, Europe 4, North America and East Asia (ex-China) 3 each; South Asia, sub-Saharan Africa and MENA one apiece, and all three of those are the day's cohort or satellite-evaluation records rather than instrumentation.

Life-course windows addressed today (records may span several stages)

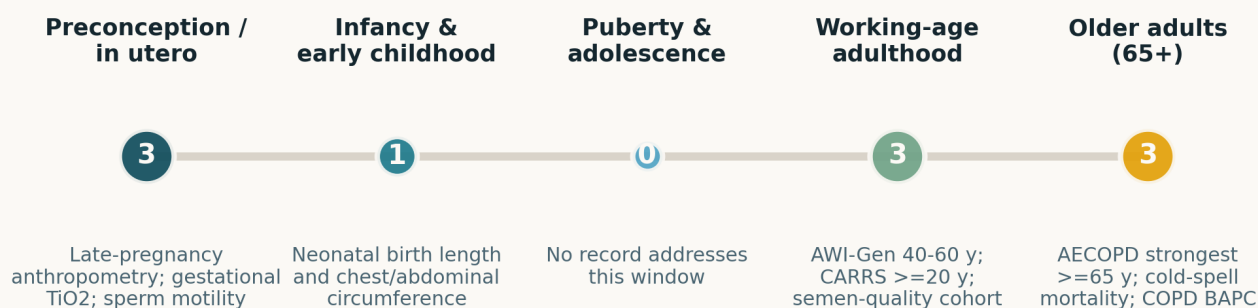


Fig. Life-course windows addressed. Counts are non-exclusive and describe this issue only. Preconception/in-utero holds three records — its highest count in eight issues (late-pregnancy anthropometry, gestational TiO₂ inhalation, and a semen-quality cohort whose exposure window is preconceptional). Puberty and adolescence is empty, as on three of the previous five issues.

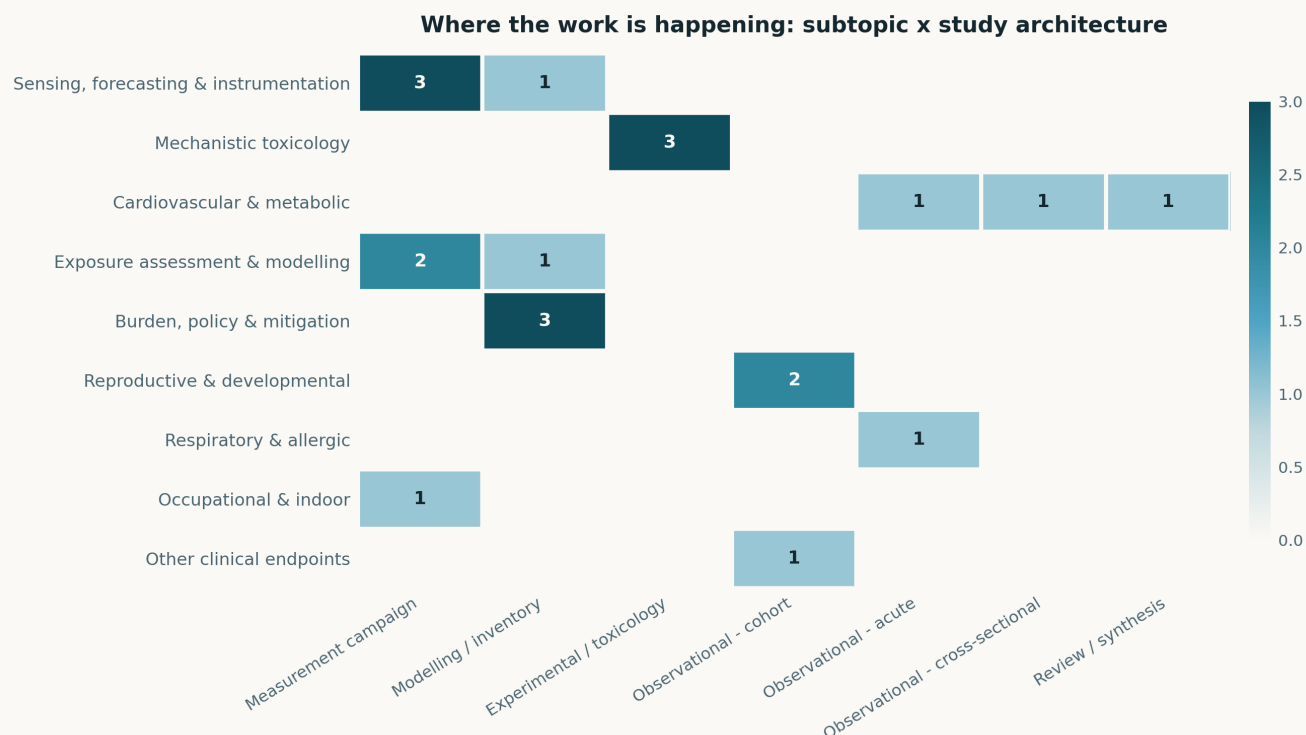


Fig. Subtopic × study-architecture cross-tabulation. Twenty-one records occupy 13 of the 70 available cells and no cell holds more than 3. The silo is visible as a clean block structure rather than a diagonal: *Sensing* and *Exposure* sit entirely in the *Measurement campaign* and *Modelling / inventory* columns, *Mechanistic toxicology* entirely in *Experimental / toxicology*, and every human-endpoint subtopic in an observational column — with one exception, the *Cardiovascular* cell in *Review / synthesis*, which is the metadata-only methodological letter. Unlike 14 August, there is no record this issue in which exposure and outcome were measured by the same group on the same subjects.

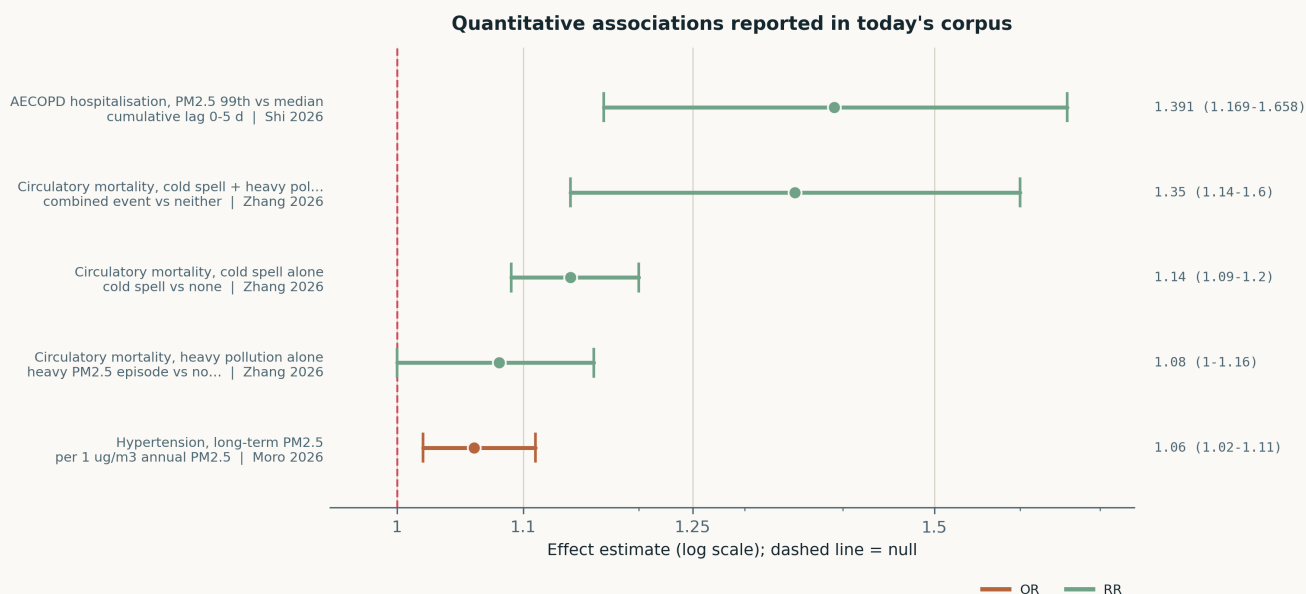


Fig. The 5 interval estimates carried today, on a common logarithmic axis — the thinnest forest plot of the week, because most of today's corpus is instrumentation. **The three Zhang et al. estimates are not independent of one another:** they are nested contrasts from the same 70-district meta-analysis, and the combined-exposure estimate shares person-time with both single-exposure estimates. Shi et al. report a 99th-percentile-versus-median contrast at cumulative lag 0–5 d, and Moro et al. a per-1 $\mu\text{g}/\text{m}^3$ odds ratio, so the exposure contrasts differ by more than the plot can show. Mandal et al.'s eGFR slopes are on an additive ml/min/1.73 m² scale and cannot be placed on a log ratio axis; they are reported in the digest and the signal box instead.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

4 records

Cho et al. 2026 Sci Total Environ

Greenspace, land use and PM_{2.5} across pandemic phases: a community scientist-led mobile sensing campaign

Community scientists ran low-cost mobile PM_{2.5} monitoring across Norwood, Ohio, for two summer periods — the initial pandemic phase (2020) and the reopening phase (2021) — with a meteorology-adjusted difference-in-differences framework for the year contrast and street-level linear land-use regression for spatial structure. **Higher 2021 concentrations were not fully explained by meteorology.** In the 2021 afternoon model, PM_{2.5} was elevated within 50 m of office and manufacturing land use and inversely associated with NDVI within 100 m. **Cross-validated R^2 0.17–0.25 (2020) rising to 0.55–0.61 (2021).** The paper attributes the gap to source recovery, but a low-contrast summer is an equally good explanation and is not tested. Health-survey scores by ward are reported alongside but are not linked to measured exposure at the individual level. The transferable part is the QC and sampling protocol for neighbourhoods with no regulatory coverage, which is stated explicitly as the deliverable.

doi: 10.1016/j.scitotenv.2026.182124

Alexa et al. 2026 Talanta

Simultaneous continuous determination of gaseous HNO₃ and particulate nitrate by chemiluminescence

Gaseous HNO₃ is sampled by wet diffusion denuder and particulate NO₃[−] in parallel by continuous aerosol sampler, both analysed online by continuous flow analysers with chemiluminescence detection — no preconcentration step, signal recorded at 1 s intervals. Detection limits (3S/N) are **5.1 nM and 12.8 nM in solution, corresponding to 0.13 µg/m³ (49.7 ppt) for HNO₃ and 0.10 µg/m³ for NO₃[−].** Applied to urban air in Brno across winter and summer campaigns 2022–2023 and compared against filter sampling with NaF-coated dry denuders. The gas–particle partitioning of nitrate is temperature- and humidity-driven on timescales of minutes, which filter-based methods average away entirely and optical low-cost sensors cannot resolve at all; a method at this time resolution is the reference any nitrate-aware correction scheme would need. The abstract reports no field intercomparison statistics against the reference method, only that the comparison was made.

doi: 10.1016/j.talanta.2026.130457

Kafi et al. 2026 Sci Total Environ

Evaluating the suitability of Sentinel-5P data for air quality monitoring in Tehran

Six years (2019–2024) of TROPOMI Level-2 CO, NO₂, SO₂, O₃ and UV Aerosol Index against 22 ground stations in a megacity with complex topography, assessed by Pearson and Spearman correlation and Kling–Gupta Efficiency, plus a spatiotemporal trajectory comparison. **Only NO₂ showed strong agreement; CO, SO₂ and O₃ were weak,** and the trajectory analysis diverged as well. The authors are explicit that UVAI represents column aerosol loading rather than surface particulate concentration and warn against reading it as PM — a caveat routinely ignored in the satellite-PM literature and worth citing for that reason alone. The diagnosis (coarse pixel size against a topographically constrained urban basin) is plausible but not separated from the competing explanation of station siting, and no vertical-profile or boundary-layer-height information is brought in to test it.

doi: 10.1016/j.scitotenv.2026.182069

Aparicio et al. 2026 Sensors (Basel)

Sensor system design and deployment for environmental and visitor monitoring in cultural heritage sites

Design, build and field validation of low-power low-cost nodes across five pilot sites in four European countries under the EU ARGUS project, in two configurations: static nodes at fixed positions and dynamic nodes carried by a quadruped robot or drone. Payload spans thermal–humidity probes, volumetric water content, wind, pollution monitors and two generations of visitor-counting devices, with Baltanás used as the iterative prototyping site before wider deployment. Carried for the mobile-platform architecture — a sensor-on-robot deployment pattern that is directly transferable to indoor and semi-enclosed PM mapping — rather than for any air-quality result. **The abstract reports no PM performance figures, no reference co-location and no calibration procedure,** which is the whole question for a low-cost node, so nothing here constrains data quality. Tier C on that basis.

doi: 10.3390/s26154941

Exposure assessment & modelling

3 records

Chebaicheb et al. 2026 Environ Int

A phenomenology of submicron organic aerosol components in urban France

The first nationwide harmonised organic-aerosol source apportionment for France: time-resolved Aerosol Chemical Speciation Monitor measurements at 12 (sub)urban sites over one to seven years each, with a single rolling PMF protocol applied under common constraints and acceptance criteria across all datasets. Primary factors — hydrocarbon-like, biomass-burning and cooking-like OA — account for **≈33% of OA mass**; oxygenated OA dominates at every site and splits into less- and more-oxidised components everywhere except Poitiers and Strasbourg. Site-specific factors resolved a mixed shipping/industrial component at Marseille Longchamp and an amine-related component at Strasbourg and Creil. Wintertime LO-OOA correlates with BBOA, implicating residential combustion in secondary as well as primary mass; summer LO-OOA tracks biogenic precursors. **CHIMERE shows systematic component-level bias against the observations.** The harmonisation is the contribution — rolling PMF applied inconsistently is why the existing European OA literature does not pool — and the limitation is that (sub)urban sites only cannot constrain the rural background these factors are transported through.

doi: 10.1016/j.envint.2026.110453

Sui et al. 2026 J Environ Manage*Contrasting annual variation of organic and elemental carbon in the Pearl River Delta*

Three years of continuous OC/EC observation (2019–2021) spanning the COVID-19 lockdown, with PMF source apportionment. OC declined steadily, driven by suppressed secondary organic carbon formation in the PRD's NO_x-rich regime — a relationship the authors report as *reversing* during lockdown when NO_x fell sharply, which is the interesting result and the one that identifies the regime. EC was flat or slightly rising, so fossil combustion from industry and mobile sources has not responded to the control programme. **Fossil-related emissions are ≈70% of carbonaceous aerosol** and that share did not fall over the period. A single-site three-year record cannot separate emission change from meteorological variability, and no such adjustment is described; the lockdown period is doing most of the causal work and is a one-off natural experiment with a confounded meteorology.

doi: 10.1016/j.jenvman.2026.130708

You et al. 2026 Environ Pollut*Satellite-based estimation of ambient ammonia and emission inventory uncertainty in South Korea*

Monthly ground-level NH₃ at 15 km resolution for 2013–2017, from Cross-track Infrared Sounder retrievals refined by a two-stage framework: a linear mixed-effects model carrying meteorology plus the Clean Air Policy Support System inventory, then a generalised additive model over the LMM residuals. **Cross-validated $R^2 = 0.73$, MAE 0.15, RMSE 0.21**. Concentrations peak in agricultural areas March–June; temperature is the dominant meteorological control (**+18.9% per +1 °C in June**), with relative humidity and wind speed strongest in March. The GAM residual surface identifies agricultural hotspots the inventory does not contain, which is the paper's actual claim — inventory gaps, not model skill. NH₃ matters here as the limiting precursor for ammonium nitrate and sulfate, so a 15 km monthly NH₃ field is a plausible predictor for secondary PM_{2.5} models, but the resolution is far coarser than the sources, and validation against sparse ground NH₃ is circular with the sparsity that motivated the work.

doi: 10.1016/j.envpol.2026.128903

Mechanistic toxicology**3 records****Mahmutovic Persson et al. 2026** Environ Res*Chronic aerosolized PM_{2.5}, plasma proteins and coronary S1P response in rats*

Well-characterised urban PM_{2.5} was re-aerosolised and delivered to rats under two regimes designed to separate an acute high dose from a chronic lower dose at comparable cumulative burden, with coronary vasomotor tone measured ex vivo by wire myograph and systemic effects read from plasma sphingosine-1-phosphate and endothelin-1 plus immunohistochemistry. **Only the chronic regime moved anything**: altered S1P vasomotor response in coronary arteries and significantly decreased plasma total protein and ET-1. The design is the point — most PM inhalation toxicology uses acute high-dose exposure because it is tractable, and this says that route can miss the effect entirely. Re-aerosolised collected PM is not fresh ambient aerosol (semivolatiles are lost, morphology changes), the ET-1 *decrease* is not obviously in the direction the vascular-dysfunction hypothesis predicts and is not reconciled, and no n or dose figures appear in the abstract.

doi: 10.1016/j.envres.2026.125391

Seymore et al. 2026 Toxicol Appl Pharmacol*Gestational nanoparticle inhalation alters placental glucose transporters and increases glucose flux*

Pregnant Sprague Dawley rats exposed by whole-body inhalation to TiO₂ nanoparticles at **9.8 ± 1.0 mg/m³**, an occupational-level concentration, from gestational day 5 to 19; GLUT1/3/4 assayed in term placenta at GD20 and ex vivo placental perfusion used as a functional test. Glut3 mRNA and GLUT1 protein fell, but GLUT4 expression and membrane localisation of both GLUT1 and GLUT4 rose, and perfusion confirmed **increased glucose flux** (AUC 95% CI 77.4–127.5 exposed vs 39.1–73.6 control). The authors state plainly that this contradicts their hypothesis. Read as a compensatory adaptation rather than a benign finding — upregulated placental glucose delivery is not obviously protective in a fetal-growth-restriction model. The dose is roughly three orders of magnitude above ambient PM_{2.5} and the particle is a single engineered material, so this constrains occupational nanomaterial exposure and says little directly about ambient PM.

doi: 10.1016/j.taap.2026.117996

Chen et al. 2026 Allergy*Gut microbiota signatures reflect PM_{2.5}-exacerbated asthma in a murine model*

Metadata-only: a research letter deposited without an abstract, so nothing below the title is verifiable from the record. Registered here because the gut–lung axis in PM-exacerbated allergic airway disease is a live mechanistic question and because the author group (Academia Sinica Air Quality Research Center with NTU clinical immunology) pairs characterised ambient aerosol with a murine model, which is unusual. No finding, effect size or exposure characterisation is claimed by this issue, and the record supports no figure or statistic in the analytics. Flagged for retrieval once the full text is indexed.

doi: 10.1111/all.70485

Cardiovascular & metabolic**3 records****Zhang et al. 2026** Zhonghua Yu Fang Yi Xue Za Zhi*Combined cold-spell and heavy-pollution exposure and circulatory mortality, Beijing–Tianjin–Hebei*

Daily mortality, PM_{2.5} and meteorology from 70 districts and counties, cold seasons (November–March) 2013–2018, mean daily PM_{2.5} **98.58 ± 78.19 μg/m³**. Three combined-exposure scenarios were constructed — overlapping cold spell and heavy pollution (A), a standalone pollution event within 7 d of a cold spell ending (B), and a standalone cold spell within 7 d of a pollution event ending (C) — with district-level DLNMs pooled by random-effects meta-analysis. The overlapping event gave **CRR 1.35 (1.14–1.60)** against **1.14 (1.09–1.20)** for cold spell alone and **1.08 (1.00–1.16)** for heavy pollution alone. Supra-additivity is the substantive result, and it is the case for joint rather than single-hazard alerting. Heavy pollution is defined by a concentration threshold, not by composition, so the interaction could equally be a composition effect: winter heating episodes in this region differ chemically from other high-mass days, and the design cannot separate the two.

doi: 10.3760/cma.j.cn112150-20260105-00006

Moro et al. 2026 Environ Res*Long-term PM_{2.5}, hypertension and metabolic syndrome in rural Ghana*

Cross-sectional analysis of 1,833 AWI-Gen participants aged 40–60 in rural Ghana, with 2007–2015 mean PM_{2.5} from a satellite-based model and MetS by harmonised criteria, under sequentially adjusted multivariable logistic regression. **OR 1.06 (1.02–1.11) per 1 µg/m³ for hypertension**, a monotone quartile gradient ($p_{\text{trend}} = 0.003$) with the top quartile at 50% greater odds, and **PAF 18.6%**. **No association with MetS overall or with any other component** — the null is as informative as the positive, and the authors report it rather than burying it. Three limitations bound the inference: the design is cross-sectional so temporality is not established; exposure is modelled at satellite resolution over a region with almost no reference monitoring and no local validation is reported; and household air pollution from solid-fuel cooking, which dominates personal exposure in this setting, is not in the model at all.

doi: 10.1016/j.envres.2026.125484

Ting 2026 JACC Adv*Statistical fragility and methodological constraints in metabolomic mediation of air pollution and cardiovascular risk*

Metadata-only: correspondence deposited without an abstract. Registered because it completes an exchange this watch is already carrying — the authors' reply (10.1016/j.jacadv.2026.103149) surfaced in the 12 August window as an Elsevier metadata-only record and remains unresolved. The subject, statistical fragility of mediation estimates in air-pollution metabolomics, is directly relevant: mediation analysis on high-dimensional metabolite panels with modest cohort sizes produces indirect effects whose confidence intervals are dominated by the product-of-coefficients variance, and the literature routinely reports them as though they were primary. No numeric claim is made from this record. Retrieve both letters together once full text is available.

doi: 10.1016/j.jacadv.2026.103150

Burden, policy & mitigation**3 records****Liu M. et al. 2026** Environ Monit Assess*Divergent near-surface ozone and NO₂ trends and ozone-attributable burden, Huaihe River Basin*

Twelve years (2013–2024) of high-resolution satellite-derived surface MDA8 O₃ and NO₂ across the Huaihe Basin, with pixel-wise OLS trends, the Hurst exponent as a persistence diagnostic, and BenMAP-CE for attributable premature mortality and economic loss. **O₃ rose at 2.792 µg m⁻³ yr⁻¹ (mean 104.15 µg m⁻³) while NO₂ fell at -1.385 µg m⁻³ yr⁻¹ (mean 29.86)** — the textbook VOC-limited response to NOx-only control, here quantified at basin scale with an outward expansion of O₃-affected area and contraction of NO₂ hotspots. Cardiovascular mortality was the largest cause-specific component, concentrated in Jiangsu and Shandong. **No PM metric is estimated**, which for a basin whose PM_{2.5} and O₃ share precursors is a real gap; and the Hurst exponent applied to a 12-point annual series is a weak persistence estimator being asked to carry a forward-looking claim.

doi: 10.1007/s10661-026-15802-3

Mou et al. 2026 Environ Pollut*Ship emissions and regional air quality in the post-DECA era, Yangtze River Basin*

A bottom-up 2023 emission inventory for **18,478 vessels** built from **≈230 million AIS trajectories** on the world's busiest inland shipping corridor, coupled to WRF-Chem for concentration attribution. SO₂ and PM_{2.5} ship emissions are substantially controlled post-DECA, but **NOx reached 156.41 kt yr⁻¹, 82.0% of major-pollutant mass**, with hotspots at Yichang–Jingzhou, Wuhan and Nanjing–Shanghai. Ship contribution to NO₂ was **0.16–3.41 µg m⁻³ annual mean** across 11 mainstem cities; PM_{2.5} increments were small but nitrate-dominated; O₃ fell in January and October under near-source NO titration and rose in July. The policy read is clean — fuel-sulfur control has done its work and the residual problem is NOx — but the inventory rests on activity-based emission factors with no plume measurement to validate them, and the PM_{2.5} increment excludes secondary nitrate formed downwind of the model domain.

doi: 10.1016/j.envpol.2026.128946

Liu D. et al. 2026 Medicine (Baltimore)*COPD burden 1990–2050 with risk-factor attribution across age and SDI*

GBD 2021 across 204 countries, analysed by joinpoint regression, age–period–cohort models and decomposition into population growth, ageing and epidemiological change, with Bayesian APC projection to 2050. Age-standardised rates fell (AAPC -0.05% incidence, -1.49% mortality, -1.48% DALYs) while absolute burden rose to **213.4 million cases and 1.69 million deaths in 2021**. **Ambient PM_{2.5} carries 15–25% of DALYs and household air pollution 0–55%**, against smoking at 38–55%, with an inverted-V SDI relationship peaking at SDI 0.4–0.6. The household-air-pollution range spanning zero to fifty-five percent is the whole story and is presented as a single interval; it is a statement about between-region heterogeneity, not uncertainty, and the paper does not separate the two. GBD risk-factor PAFs inherit their exposure–response functions from the integrated exposure–response curves, so this is a re-analysis of GBD outputs rather than independent evidence. Tier C accordingly.

doi: 10.1097/MD.00000000000050162

Reproductive & developmental**2 records****Hung et al. 2026** iScience*Late-pregnancy air pollution, gestational hypertension and neonatal anthropometry*

Women recruited in the second trimester and classified gestational-hypertensive or normotensive, with PM_{2.5}, PM₁₀, NO₂, SO₂, CO and O₃ over the 30 days before delivery assigned from ambient monitors, and maternal peridelivery hs-CRP, IL-1β and TNF-α plus cord TNF-α as candidate mediators. GH women had higher hs-CRP and delivered lighter neonates with smaller chest and abdominal circumferences; PM, NO₂, SO₂ and CO were **inversely associated with birth length**. **Mediation analysis found no significant indirect effect through any measured cytokine**, which is the useful negative — the inflammatory-mediation hypothesis is the default assumption in this literature and it did not hold here. Ambient monitor assignment over a 30-day window is coarse for a length endpoint measured in centimetres, no sample size or effect sizes appear in the abstract, and a single peridelivery cytokine draw is a poor proxy for a gestation-long inflammatory exposure.

doi: 10.1016/j.isci.2026.117029

Wang et al. 2026 Environ Pollut*PM_{2.5} modifies the temperature–semen quality association*

Prospective cohort with repeated semen analysis, linear mixed models with natural cubic splines for temperature, then piecewise regression stratified into higher and lower PM_{2.5} groups across the low- and high-temperature segments. Temperature showed **inverted-U associations with semen quality over the 0–90, 0–9 and 10–14 d windows** — 0–90 d spans a full spermatogenic cycle, 10–14 d the epididymal transit, so the window structure is biologically coherent. **Synergy between high temperature and PM_{2.5}, antagonism between low temperature and PM_{2.5}**, both acting mainly on progressive and total motility. Stratifying at a PM_{2.5} median is a crude interaction test that discards within-stratum contrast, and antagonism at low temperature is at least as easily explained by seasonal confounding (heating-season composition, behaviour, indoor time) as by a joint biological mechanism. No effect sizes or intervals are reported in the abstract, so nothing from this record enters the forest plot.

doi: 10.1016/j.envpol.2026.128910

Respiratory & allergic**1 record****Shi et al. 2026** Medicine (Baltimore)*Air pollution and AECOPD hospitalisation across three Hebei cities*

Ecological time series on **14,786 AECOPD hospitalisations** from 49 hospitals in Shijiazhuang, Handan and Chengde, 2018–2020 (1,096 days), with DLNMs adjusted for long-term trend, temperature, humidity, day of week, holidays and pandemic stage, then pooled by random-effects multivariate meta-analysis. Cumulative effects strengthened with lag length and peaked at lag 0–5 d for every pollutant while single-day effects were weak. **PM_{2.5} at the 99th percentile versus median gave RR 1.391 (1.169–1.658) at lag 0–5**; all five non-ozone pollutants showed non-linear cumulative exposure–response, and higher 8-h ozone was associated with *lower* admission risk, which the authors do not explain and which is most plausibly seasonal confounding. The abstract's own conclusions are internally inconsistent on susceptibility — the results state associations were more pronounced in males, the conclusion says females — so the stratified findings should not be cited without the full text. Six correlated pollutants tested across six lag structures with no multiplicity adjustment.

doi: 10.1097/MD.0000000000005080

Occupational & indoor exposure**1 record****Ranjbar et al. 2026** Preprints.org (preprint)*1-Nitropyrene as a diesel-specific indoor exposure proxy in California homes*

1-Nitropyrene and the isomeric 2-nitropyrene and 2-nitrofluoranthene measured in indoor air and house dust from **40 homes** in the eastern San Francisco Bay Area across two sampling events, alongside continuous indoor black carbon. The motivation is specificity: BC, NO_x and PM_{2.5} are all used as diesel surrogates and none is diesel-specific. Detection was **74% in air and 97% in dust**, medians **0.42 pg/m³ (IQR 0.32–0.54)** and **340 pg/g (IQR 190–620)**; air–dust correlation $r = 0.40$ ($p = 0.02$) and BC–1-NP $r = 0.38$ (air) and 0.47 (dust). Diagnostic ratios (2-NF1:1-NP < 5) indicate primary emission rather than secondary formation. Dust 1-NP associated with more spatial diesel predictors than air 1-NP did, consistent with dust integrating longer-term deposition — which is the actual argument for using it. Preprint, not peer reviewed. Forty homes, two events, no personal sampling, and a picogram-per-cubic-metre analyte whose blank and recovery performance is not summarised in the abstract.

doi: 10.20944/preprints202608.0873.v1

Other clinical endpoints**1 record****Mandal et al. 2026** Kidney Int Rep*Long-term ambient PM_{2.5} and kidney function in urban Indian adults*

Population-representative cohort of **12,271 adults** aged 20 and over in Chennai and Delhi, with eGFR (CKD-EPI 2009) at three points between 2010 and 2016 and annual mean PM_{2.5} assigned at residential geocode from a 1 km national daily spatiotemporal model. City-specific linear mixed models with inverse-probability weighting for loss to follow-up, plus effect modification by sex, waist–hip ratio, hypertension and diabetes. Exposure contrast between cities is extreme — Chennai 33/37/30 versus Delhi 118/123/130 $\mu\text{g}/\text{m}^3$ — and both cities show decline: **–0.32 (95% CI –0.49 to –0.15)** in Chennai and **–0.42 (–0.57 to –0.27) ml/min/1.73 m² per 5 $\mu\text{g}/\text{m}^3$** in Delhi. That the two cities agree despite a fourfold concentration difference is reassuring for the shape of the relationship and awkward for its magnitude. CKD-EPI 2009 is poorly calibrated in South Asian populations, three measurements over six years is thin for an eGFR slope, and the by-city stratification means the between-city contrast — the largest exposure contrast in the data — is never used.

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Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Moro et al.	Long-term ambient PM _{2.5} exposure, hypertension and metabolic syndrome among Ghanaian adults	<i>Environ Res</i>	Cross-sectional (cohort baseline)	PM _{2.5} (satellite-based model, 2007-2015 mean)	Ghana
Ting	Statistical fragility and methodological constraints in metabolomic mediation of air pollution and cardiovascular risk	<i>JACC Adv</i>	Method review	Ambient PM (methodological commentary)	Not stated
Zhang et al.	Acute impact of combined exposure to cold spells and heavy pollution on circulatory disease mortality in the Beijing-Tianjin-Hebei region	<i>Zhonghua Yu Fang Yi Xue Za Zhi</i>	Distributed-lag time-series	PM _{2.5} , heavy-pollution thresholds >75 / >115 / >150 µg/m ³	China
Reproductive & developmental					
Hung et al.	Late-pregnancy air pollution exposure and neonatal anthropometry with and without gestational hypertension	<i>iScience</i>	Prospective birth cohort	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO, O ₃ (30 d pre-delivery, ambient monitors)	Taiwan
Wang et al.	Synergistic and antagonistic interactions of PM _{2.5} with high and low temperatures on sperm motility	<i>Environ Pollut</i>	Observational - cohort	PM _{2.5} with temperature, 0-90 / 0-9 / 10-14 d exposure windows	China
Respiratory & allergic					
Shi et al.	Air pollution and acute exacerbation of COPD in Hebei Province: a multi-city time-series study	<i>Medicine (Baltimore)</i>	Distributed-lag time-series	PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ , CO, 8-h O ₃	China
Mechanistic toxicology					
Chen et al.	Gut microbiota signatures reflect particulate matter 2.5-exacerbated asthma in a murine model	<i>Allergy</i>	Animal model	PM _{2.5} (murine asthma model)	Taiwan
Mahmutovic Persson et al.	Chronic exposure to aerosolized PM _{2.5} in rats alters plasma protein levels and ex vivo coronary artery response to sphingosine-1-phosphate	<i>Environ Res</i>	Animal model	Re-aerosolized urban PM _{2.5} (acute high-dose vs chronic low-dose)	Europe
Seymore et al.	Gestational inhalation of nanoparticles alters placental glucose transporter expression and increases glucose flux in rats	<i>Toxicol Appl Pharmacol</i>	Animal model	TiO ₂ nanoparticles, 9.8 +/- 1.0 mg/m ³ whole-body inhalation	USA
Occupational & indoor					
Ranjbar et al.	Evaluating diesel exhaust exposure in California homes using 1-nitropyrene in indoor air and dust	<i>Preprints.org (preprint)</i>	Measurement campaign	1-Nitropyrene, 2-nitropyrene, 2-nitrofluoranthene; continuous indoor black carbon	USA
Sensing, forecasting & instrumentation					
Alexa et al.	Simultaneous determination of gaseous nitric acid and particulate nitrate in ambient air using continuous samplers and chemiluminescent online analysers	<i>Talanta</i>	Multi-instrument field evaluation	Gaseous HNO ₃ and particulate NO ₃ -, 1 s chemiluminescence	Czech Republic
Aparicio et al.	Sensor system design and deployment for environmental and visitor monitoring in cultural heritage sites	<i>Sensors (Basel)</i>	Network deployment + field evaluation	Low-cost pollution monitors + T/RH, VWC, wind; static and robot/drone-borne	Europe
Cho et al.	Greenspace, land use and PM _{2.5} across pandemic phases: a community scientist-led mobile sensing campaign	<i>Sci Total Environ</i>	Mobile monitoring campaign	PM _{2.5} (low-cost mobile sensors, street-level)	USA
Kafi et al.	Evaluating the suitability of Sentinel-5P satellite data for air quality monitoring in Tehran	<i>Sci Total Environ</i>	Satellite retrieval evaluation	TROPOMI CO, NO ₂ , SO ₂ , O ₃ and UV Aerosol Index (no direct PM product)	Iran
Exposure assessment & modelling					
Chebaicheb et al.	A phenomenology of submicron organic aerosol components in urban France from long-term online observations and modelling	<i>Environ Int</i>	Source apportionment	PM ₁ organic aerosol, ACSM, rolling PMF (HOA/BBOA/COA/LO-OOA/MO-OOA)	France
Sui et al.	Contrasting annual variation of organic and elemental carbon in the Pearl River Delta across the COVID-19 lockdown	<i>J Environ Manage</i>	Source apportionment	PM _{2.5} organic and elemental carbon, PMF	China

First author	Focus	Journal	Design	Metric	Region
Yeu et al.	Satellite-based estimation of ambient ammonia reveals meteorological influences and emission inventory uncertainties in South Korea	<i>Environ Pollut</i>	Satellite remote sensing	Ground-level NH ₃ (secondary PM _{2.5} precursor), CrIS, 15 km monthly	South Korea
Burden, policy & mitigation					
Liu D. et al.	COPD burden 1990-2050: joinpoint, age-period-cohort and Bayesian APC modelling of diverging risk-factor attribution	<i>Medicine (Baltimore)</i>	Burden estimation (GBD)	Ambient PM _{2.5} and household air pollution as GBD risk factors (PAF)	Global
Liu M. et al.	Divergent trends in near-surface ozone and nitrogen dioxide and ozone-attributable health burden in the Huaihe River Basin, 2013-2024	<i>Environ Monit Assess</i>	Satellite remote sensing	Satellite-derived surface MDA8 O ₃ and NO ₂ (no PM metric)	China
Mou et al.	Impacts of ship emissions on regional air quality in the post-DECA era: the Yangtze River Basin	<i>Environ Pollut</i>	CTM simulation + source attribution	Ship-source PM _{2.5} , SO ₂ , NO _x , nitrate; WRF-Chem increments	China
Other clinical endpoints					
Mandal et al.	Long-term exposure to ambient particulate matter and kidney function in urban Indian adults	<i>Kidney Int Rep</i>	Observational - cohort	Annual mean PM _{2.5} , 1 km x 1 km national spatiotemporal model at residential geocode	India

Provenance and method

Window. PubMed [EDAT] 2026-08-15 to 2026-08-15, contiguous with the 14 August issue; no gap. Every record was checked against `state/seen.json` (pmid → lowercased doi → title signature) before summarising; **19 Europe PMC hits were dropped as already-seen duplicates** of records carried on 8–14 August, the largest cross-source collision so far.

Per-leg yield. PubMed connector, two separate queries on the health and instrumentation axes: the health axis returned 7 hits and the instrumentation axis, once narrowed to exclude the `calibration/exposure` model MeSH explosions that returned 76 mostly-clinical records, returned 1 — fully contained in the health set. **6 carried** from the connector. Local harvester: `pubmed_health` 5, all contained in the connector set; `pubmed_sensing` 0. **Europe PMC carried the issue:** 59 hits on `CREATION_DATE`, 40 new after deduplication, **15 carried** — including every instrumentation record in this digest. Crossref by ISSN returned 9 across 18 journals, of which one (*Air Qual Atmos Health*, North African urban anomalies) was in scope but arrived after screening closed and is deferred to the weekly sweep; the rest were *Environ Sci Technol* and *Environ Monit Assess* records outside scope or already carried. arXiv 146 kB, nothing on or after 11 August. **OpenAlex HTTP 429, 13th consecutive** — the leg is dead and no further effort is warranted. ClinicalTrials.gov: no PM or air-pollution registration created or updated inside the window; `trials.json` unchanged.

Harvester defect found. `harvest.py`'s Europe PMC leg returned 0 for this window while a hand-issued `CREATION_DATE: [2026-08-15 TO 2026-08-15]` query against the same endpoint returned 59. Fifteen of today's 21 records came from the manual query, so the committed leg would have shipped a six-record issue. Logged to `PIPELINE.md` for repair before the next run; the daily harvest cannot be trusted on the Europe PMC leg until it is fixed.

Rejections (26). By reason: eleven keyword collisions where `aerosol` or `air pollution` matched an unrelated field (PIPAC oncology, therapeutic aerosol in equine wound care, free-space optical communication, solar UV dosimetry, veterinary virology, CF pharmacology, infection-control adherence, broiler LCA, Nile Red photophysics, X-ray thermometry of glycerol droplets, a thermoelectric gas sensor with no particulate channel); four pharmaceutical-inhaler records; three emissions/chemistry papers with no PM measurement (NO_x budget, brominated dioxins, green nanomaterials); a VOC-only vehicle catalyst study framed as PM-precursor work but reporting no PM; a scoping review whose literature search closed in April 2021; an allergic-rhinitis DLNM carrying meteorology only, with pollution as an adjustment covariate; a stakeholder-perception window-design study with no measured indoor PM; a veterinary climate-health commentary; a duplicate methodological correspondence; and one PMC front-matter item. All logged with reasons to `state/rejected.jsonl`.

Two records are metadata-only (Chen et al., *Allergy*; Ting, *JACC Adv*) — deposited without abstracts, registered by title only, and supporting no numeric claim anywhere in this issue. **One record is a preprint** (Ranjbar et al., Preprints.org) and is labelled as such in the register.

Labels. Subtopic, design, particle-metric and geography labels were drawn only from vocabularies already mapped in `plots.py` and `mktable.py`; no new leaf label was written this issue. Assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. `check_dois.py` returned 0 fail, 0 warn across all 21 records.

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